Silicon-Organic Hybrid (SOH) Racetrack Modulators for 200 Gbit/s PAM4 Signaling with Ultra-low Drive Voltages

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Abstract: We demonstrate a silicon-organic hybrid (SOH) racetrack modulator enabling 200 Gbit/s PAM4 signaling at drive voltages with a 220 mV_{pp} peak-to-peak swing – a record-low value for technically relevant data rates. Our experiment represents the first demonstration of optical signaling in an SOH racetrack modulator. © 2025 The Author(s)

1. Introduction

Massively parallel processing in hyperscale data centers and artificial-intelligence (AI) clusters continues to drive the need for faster and faster optical interconnects that can seamlessly interface with CMOS electronics. A promising path towards power-efficient high-throughput links is to use large numbers of parallel channels at rather moderate speeds. These so-called wide-and-slow (WaS) architectures [1] crucially rely on efficient and compact electro-optic modulators (EOM) that can be integrated in large arrays and that are ideally amenable to operation by rather weak drive signals with peak-to-peak swings below 1 V_{pp} as offered by CMOS circuits. In this context, resonant EOM based on ring or racetrack cavities are being investigated as a particularly promising solution, also offering an interesting path towards highly scalable wavelength-division multiplexing (WDM) schemes [2-7]. In the past, silicon photonic (SiP) micro-ring modulators (MRM) have been shown with impressive line rates of 290 Gbit/s using six-level pulse-amplitude modulation (PAM6) at peak-to-peak drive voltages of 400 mV_{pp} [2]. However, the underlying depletion-type SiP phase shifters were subject to rather limited shifting efficiencies of only 3 GHz/V which required an operating point close to the minimum of the resonant transmission dip, leading to a rather high modulation loss of around 6 dB. High phase shifting efficiencies in excess of 40 GHz/V can be achieved through forward-biased injection-type *p-i-n* phase shifters [3], but transmission was limited to 20 Gbit/s and rather simple on-off-keying (OOK). Moreover, both depletion- and injection-type SiP MRM have to rely on power-hungry thermal phase shifters for tuning to a given operating point with respect to a fixed-frequency laser [8]. In contrast, resonant EOM based on the Pockels effect offer higher shifting efficiencies while enabling power-efficient bias-point control via a DC voltage that is not associated with any current. Pockels-type racetrack modulators (RTM) have been demonstrated by combining barium titanate (BTO) as an electro-optic material with plasmonic waveguide structures, leading to outstanding phase-shifter efficiencies of more than 50 GHz/V and offering OOK modulation at 200 Gbit/s [6,7]. However, the peak-to-peak drive voltages still exceeded 1 V_{pp} (root-mean-square 0.4 V_{rms}) and the device concept is not well suited for co-integration with SiP circuits, requiring adaptation of front-end-of-line fabrication processes. Plasmonic-organic hybrid (POH) RTM also reach impressive shifting efficiencies of 22 GHz/V and allow for ultra-high line rates of up to 408 Gbit/s [4]. However, these devices suffer from rather high losses and limited Q-factors of only 700, requiring comparatively high drive-voltage swings beyond 1 V_{pp} [4]. There is hence an unmet need for optically resonant Pockels-type modulators that combine high shifting efficiencies with decent quality factors and that enable high-speed signaling with CMOS-compatible sub-1 V peak-to-peak swings.

In this paper, we demonstrate a silicon-organic hybrid (SOH) racetrack modulator that combines a Q factor of more than 12 000 with an outstanding shifting efficiency of 25.5 GHz/V. The devices can be seamlessly co-integrated with the wealth of existing SiP circuitry without changes in front-end-of-line fabrication, thereby offering a direct path to cost-efficient mass production of highly functional transceiver chips [9-11]. We demonstrate the viability of the concept in a high-speed signaling experiment, using a bias point with only 2 dB of optical modulation loss. We show 100 GBd PAM4 signaling at line rates of 200 Gbit/s using a drive signal with peak-to-peak swing of only 220 mV_{pp}. To the best of our knowledge, this is the lowest swing so far shown for high-speed signaling in a resonant modulator at technically relevant data rates.

We further demonstrate 40 GBd PAM8 signaling with a drive-voltage swing of 100 mV_{pp}, resulting in an electrical energy dissipation of only 12.8 aJ/bit. To the best of our knowledge, this is the lowest energy dissipation so far demonstrated for signaling at line rates beyond 100 Gbit/s, irrespective of the underlying device concept. We believe that our work can pave a path to highly efficient and compact modulator arrays that can be operated by CMOS-compatible drive signals with sub-1V peak-to-peak swings, thereby building the base of future WaS link architectures.

2. SOH Racetrack Modulator

A top view schematic of the SOH RTM is shown in Fig 1(a). The electrical drive signal is applied to coplanar RF contacts (yellow). Light is coupled in and out of the silicon bus waveguide (WG) via grating couplers (GC). A 7.5 µm-long directional coupler with a 220 nm-wide gap couples light from the bus strip WG to a strip WG in the cavity. After a 90° bend of the strip WG, strip-to-slot (S2S) converters are used to couple light in and out of the two slot-WG phase-shifter sections (length L_{PS} = 86 µm). The cross-section of the slot-WG phase shifter along the dashed line A-A’ in is shown in Fig 1(b). The slot WG comprises two 160-nm wide

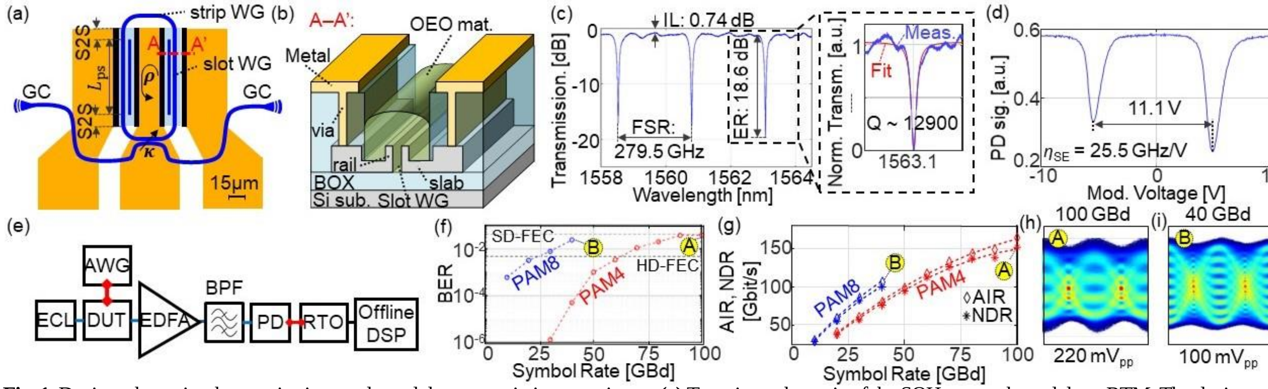

Fig. 1. Device schematic, characterization results and data transmission experiment. **(a)** Top-view schematic of the SOH racetrack modulator RTM. The device comprises coplanar RF electrodes (yellow) and optical strip and slot waveguides (WG, blue). Light is coupled in and out of the bus strip WG via grating couplers (GC) and routed into the the racetrack resonator through directional coupler with amplitude coupling coefficient κ. Strip-to-slot (S2S) converters connect strip-WG and slot-WG sections (length L_{PS} = 86 um). **(b)** Cross-section of the SOH phase-shifter WG along line A-A' in (a). The slot WG consists of two rails separated by a slot filled with OEO material. The rails are connected to the electrodes by vias and doped silicon slabs. **(c)** Optical transmission spectrum of the SOH RTM. We measure a free spectral range (FSR) of 279.5 GHz, an extinction ratio (ER) of 18.6 dB and an insertion loss (IL) of 0.75 dB. We find a loaded Q factor of 12 900 by fitting with a Lorentzian, see inset. **(d)** Optical transmission of the device vs. applied modulating voltage for a fixed laser wavelength of 1560 nm. We find a shifting efficiency of η_{SE} = 25.5 GHz/V. **(e)** Schematic of high-speed signaling setup. Light from an external-cavity laser (ECL) is coupled to the device under test (DUT). An arbitrary waveform generator (AWG) provides the electrical drive signal. At the output of the DUT, an erbium doped fiber amplifier (EDFA) amplifies the light. Out-of-band noise is suppressed by a bandpass filter (BPF). The resulting optical signal is received by a high speed photo-diode (PD) and real-time oscilloscope (RTO). An offline DSP chain is applied, and the bit error ratio (BER) and generalized mutual information are calculated. The BER as a function of symbol rate is shown in **(f)** for PAM4 and PAM8 along with the thresholds for soft-decision forward error correction (SD-FEC) and hard-decision FEC (HD-FEC) with 20 % and 7 % overhead, respectively. The net data rate (NDR) and achievable information rate (AIR) are calculated and shown in **(g)**. We find NDR of 160 Gbit/s for 100 GBd PAM4 and of 100 Gbit/s for 40 GBd PAM8. For selected data points (Ⓐ, Ⓑ) in (f) and (g), the eye diagrams as well as peak-to-peak drive voltages are provided in **(h)**–**(i)**.

silicon rails, which are connected to the metal electrodes through metal vias and doped Si slabs, the latter forming an RC-lowpass. We use an advanced multi-level doping scheme to ensure high bandwidth while maintaining low waveguide losses [12]. The base structure of the SOH RTM is fabricated in a commercial SiP foundry along with the full portfolio of standard SiP devices [9], and the organic electro-optic (OEO) material is applied in a subsequent back-end-of-line process, see [10,11] for details. Figure 1(c) shows the transmission spectrum of the SOH RTM. The off-resonance on-chip insertion loss (IL) is measured to be 0.74 dB, and the device shows a free spectral range (FSR) of 279.5 GHz and an extinction ratio (ER) of 18.6 dB. We extract a quality factor of Q ≈ 12 900 by fitting the measured transmission dip at 1563.1 nm to a Lorentzian-type model function, see inset in Fig. 1(c). Based on the Q-factor and the ER, we extract a round-trip amplitude transmission factor of ρ = 0.906 (round-trip loss ~ 0.86 dB) and an amplitude coupling coefficient κ = 0.379. After the BEOL processing, we measure the shifting efficiency η_{SE} of the SOH RTM. To this end, we fix the wavelength and measure the optical transmission through the device while sweeping the drive voltage from -10 V to 10 V. This leads to a phase shift of more than 3π such that two neighboring resonances are swept across the fixed laser line, generating two dips in the measured output power. The spacing of these dips corresponds to a phase shift of 2π, see Fig. 1(d). The shifting efficiency is obtained by dividing the FSR of the RTM, Δf_{FSR} = 279.5 GHz, by the voltage increment ΔU_d = 11.1 V required for tuning from one dip to the next, leading to an ultra-high shifting efficiency of $\eta_{SE} = \Delta f_{FS}/\Delta U_d$ = 25.5 GHz/V.

3. Data transmission experiment

To assess the performance of our SOH RTM, we conduct a high-speed signaling experiment, see Fig. 1(e) for the underlying setup. An external-cavity laser (ECL) provides the optical carrier at a power of 4 dBm and a wavelength around 1560 nm and is coupled to the device under test (DUT) via a GC (5.5 dB loss). The electrical drive signal is provided by an arbitrary-waveform generator (AWG, Keysight M8199B) and consists of a pseudo random bit sequence shaped by a root-raised cosine filter. The AWG is connected to the DUT via a 20 cm long RF cable and a RF probe. A linear minimum-mean-square-error (MMSE) equalizer is applied to the predistort the signal, compensating for the RF characteristics of the AWG and the RF cable, but not of the modulator. We adjust the operating point of our SOH RTM via wavelength tuning to the 2 dB transmission point, which slightly increases the bandwidth of our device at the expense of slope efficiency and linearity [13]. In the fiber after the DUT, we measure an optical output power of approximately -9.5 dBm for the modulated data signal. This corresponds to a substantial fiber-to-fiber loss of approximately 13.75 dB in operation, where 11 dB are caused by the rather inefficient GC. By using edge couplers and advanced packaging schemes based on 3D printed photonic wirebonds or facet-attached micro lenses instead, a significant reduction of the coupling losses down to 1.5 dB per interface is possible [14,15], potentially reducing the total fiber-to-fiber loss to 5.7 dB (including modulation loss) and increasing the output power to approximately -1.7 dBm. To compensate for the fiber-chip coupling loss, our experiment relied on an erbium doped fiber amplifier (EDFA) to boost the optical output power to a sufficient level of 8 dBm, as needed by the high-speed photo-diode (PD) and the subsequent real-time oscilloscope (RTO). The recorded signal is finally processed and demodulated by offline digital signal processing (DSP), comprising up-sampling, timing recovery, linear Sato-equalization, as well as decision-directed least-mean-square equalization, before calculating the bit error ratio (BER) and the normalized generalized mutual information (NGMI) based on a log-likelihood ratio for additive white Gaussian noise as main channel impairment [16]. During the experiment, we optimized the drive voltages for minimum BER. To evaluate the resulting the per-bit electrical energy dissipation W_{bit} of the RTM, we measure the peak-to-peak drive voltages supplied to the RTM by connecting the output of the AWG directly to the input of the RTO via the 20 cm-long RF cable. Note that we operated the SOH racetrack as a capacitive load, which leads to an effective doubling of the drive voltage at our devices. However, to stay consistent with previous literature in the field [17,18], we use the voltage swing measured for a 50 Ω load as a metric for the drive-voltage requirements of our device. To calculate W_{bit}, we furthermore need to estimate the device capacitance C_{dev} comprising both the differential slot capacitance C_{slot} as well as the RF pad capacitance C_{pad}. Based on values reported in literature [19], we estimate the total capacitance of our device to be $C_{\mathrm{dev}} = 18$ fF. The BER as a function of symbol rate is shown in Fig. 1(f) for PAM4 (○) and PAM8 (○) along with the thresholds for soft-decision (SD) forward-error correction (FEC) and hard-decision (HD) FEC with 20 % and 7 % overhead, respectively. For PAM4 we achieve a symbol rate (line rate) of 60 GBd (120 Gbit/s) with a BER below the HD-FEC limit, and 20 GBd (60 Gbit/s) were reached for PAM8. We measure a drive-voltage swing U_{d} of 140 $\mathrm{mV_{pp}}$ for PAM4 and of 110 $\mathrm{mV_{pp}}$ for PAM8, leading to an electrical energy dissipation of $W_{\mathrm{bit}} = 20/72 \times C_{\mathrm{dev}} U_{\mathrm{d}}^2 = 60.5$ aJ/bit for 60 GBd PAM4 and of $W_{\mathrm{bit}} = 4/56 \times C_{\mathrm{dev}} U_{\mathrm{d}}^2 = 25$ aJ/bit for 20 GBd PAM8 [19]. For PAM8, we increase the symbol rate (line rate) to 40 GBd (120 Gbit/s) and measure a BER of 2.4×10^{-2}. With a drive voltage of $U_{\mathrm{d}} = 100\ \mathrm{mV_{pp}}$, we calculate an energy dissipation of 13 aJ/bit, which, to the best of our knowledge, is the lowest energy dissipation so far demonstrated for a resonant electro-optic device operating at line rates above 100 Gbit/s. Similar electrical energy dissipations of 10 aJ/bit utilizing an indium phosphide MRM [5] have so far been limited to line rates of only 1 Gbit/s, whereas higher line rates of up to 290 Gbit/s with a Si MRM resulted in an electrical energy dissipation in excess of 80 aJ/bit [2]. For PAM4 at a symbol rate of 100 GBd (200 Gbit/s), we increased the output power of the laser to 6 dBm and achieve a BER of 3.8×10^{-2} with a drive voltage of only 220 $\mathrm{mV_{pp}}$ – a record-low value for signaling in resonant modulators at technically relevant data rates above 64 Gbit/s. Based on the calculated NGMI, we calculate the achievable information rate (AIR) and net data rate (NDR) based on FEC implementations reported in literature [17]. For each recording, both AIR and NDR are shown in Fig. 1(g) for PAM4 (AIR: ◊, NDR: *) and PAM8 (AIR: ◊, NDR: *). For PAM4 at 100 GBd, we find an AIR of 164.5 Gbit/s and an NDR of 152 Gbit/s. For 40 GBd PAM8, the AIR amounts to 106.5 Gbit/s, and the NDR is 100.1 Gbit/s. Note that the bandwidth of our device is currently limited by the rather large Q-factor. By reducing the Q factor, we believe even higher symbol rate should be possible, clearly at slightly increased drive voltage. Still, the high shifting efficiency of our devices should permit highly competitive performance also in this regime.

4. Summary

We demonstrate the first high-speed racetrack modulator (RTM) on the silicon-organic hybrid (SOH) platform and prove its viability in signaling experiments. Our devices combine Q factors of more than 12 000 with outstanding shifting efficiencies of more than 25 GHz/V. We show 100 GBd PAM4 signaling at line rates of 200 Gbit/s using drive signals with only 220 $\mathrm{mV_{pp}}$ peak-to-peak swing at a bias point with only 2 dB of optical loss. To the best of our knowledge, this is the lowest voltage swing so far demonstrated for high-speed signaling in a resonant modulator at technically relevant data rates. We further demonstrate 40 GBd PAM8 signaling at a line rate of 120 Gbit/s with a drive-voltage swing of 100 $\mathrm{mV_{pp}}$, resulting in an electrical energy dissipation of only 12.8 aJ/bit – a record-low value for signaling beyond 100 Gbit/s. We believe that our work paves a path towards highly efficient modulator arrays that can build the base of future wide-and-slow (WaS) optical link architectures.

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